Musicology

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Musicology

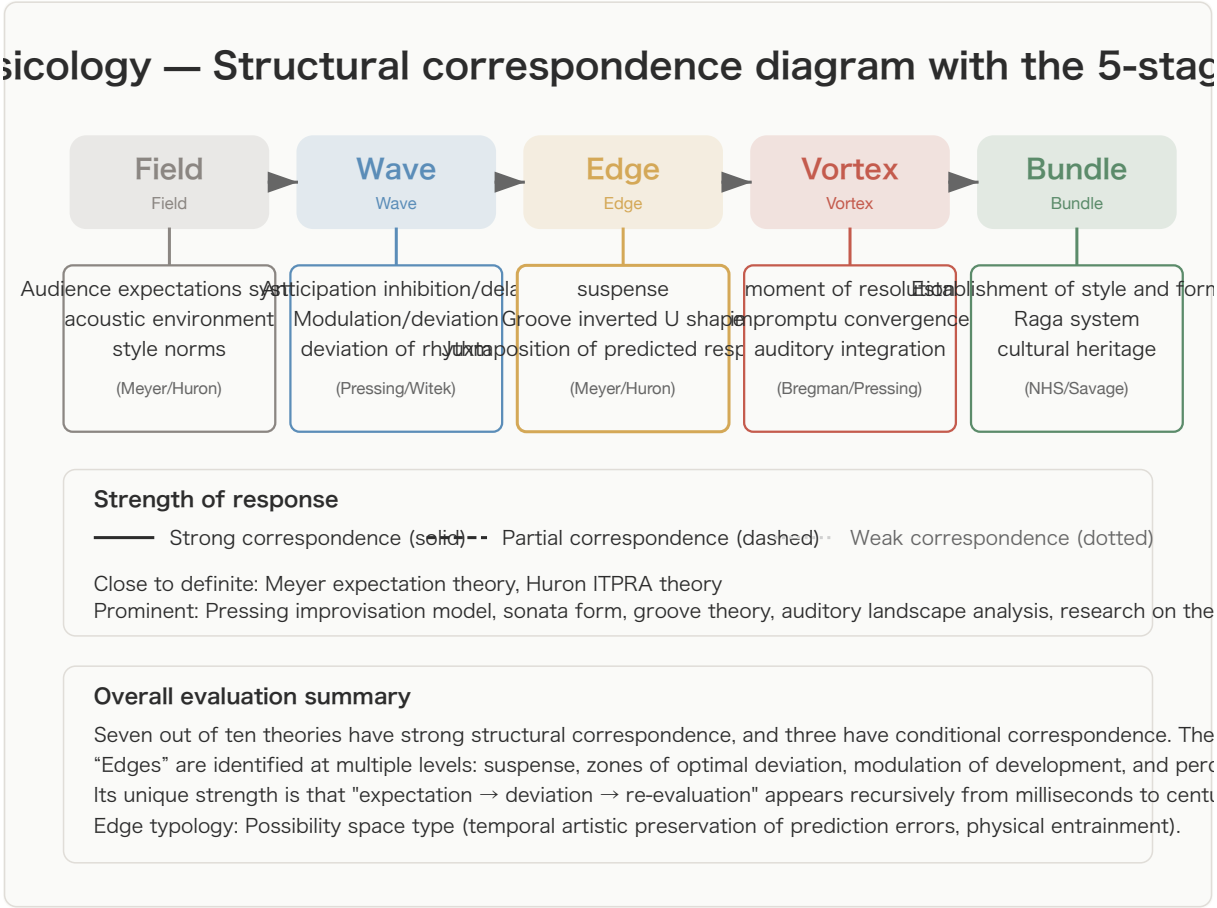

Figure 1: Musicology — Structural correspondence diagram

1. Purpose and Questions of the Investigation

This report was prepared to answer the following question:

> Do the theories in this academic domain correspond structurally to the five-stage model (Field ( ) -> Wave ( ) -> Edge ( ) -> Vortex ( ) -> Bundle ( ))?

Musicology is one of the domains that most explicitly describes the creative loop of "expectation -> error -> reevaluation." Meyer's theory of expectation is a central point of connection within the predictive-processing cluster, while Pressing's improvisational cycle and sonata form each display structural similarities to the five stages at different scales. Huron's ITPRA theory develops Meyer in computational terms, and groove theory, raga, Auditory Scene Analysis (ASA), and ecological perception complement a multidimensional understanding of musical creation.

This report presents investigation results; it is not a document written to justify any particular theory. Areas where correspondence is strong and areas where reservations are necessary are described with the same level of neutrality.

2. Method of Investigation

Scope and Selection Criteria

In this domain, 10 theories and practical models were selected for evaluation. The selection was based on three criteria: (1) explicit treatment of processes of generation or transformation; (2)

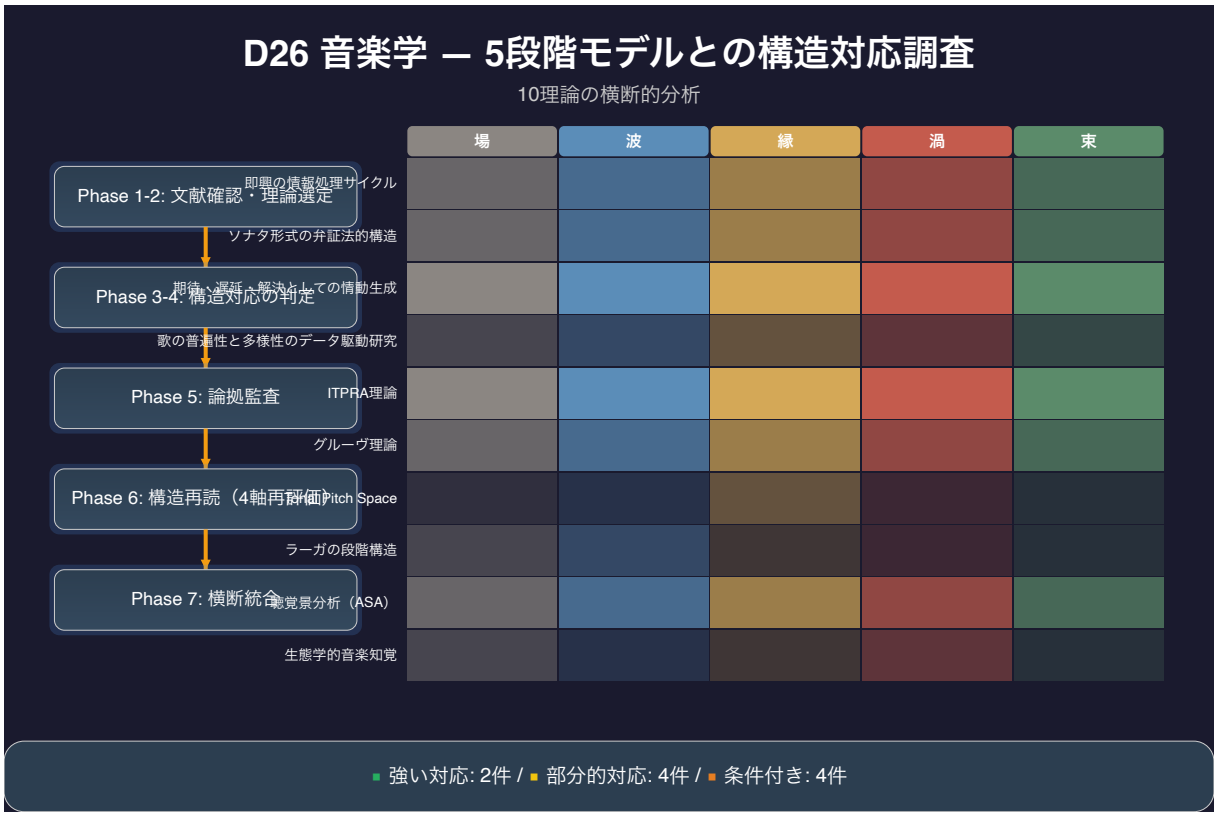

Figure 2: Musicology —

the possibility of structural analysis grounded in primary sources or well-established secondary literature; and (3) the possibility of comparing different lineages within the domain.

Evaluation Method

For each theory, the report first explains what the theory describes and then examines where correspondence with the five stages can be identified. Judgments of correspondence were made comprehensively on the basis of textual concepts, concrete examples, connections between stages, the presence or absence of circulation from Bundle to the next Field, and the risk of overinterpretation.

Criteria of Judgment

- **Strong structural correspondence**: Cases in which the core concepts of a theory correspond concretely to multiple stages of the five-stage model, and the reasons can be explained in the main text
- **Conditional correspondence**: Cases in which correspondence can be indicated, but interpretive reservations remain in the stage mapping
- **No correspondence confirmed**: Cases in which partial approximations exist, but the overall five-stage structure cannot be read without strain

Limits of the Investigation

This investigation is not an exhaustive review of every work within each theory. It focuses on the core concepts of major works and extracts structures at a level of granularity suitable for comparison. As a result, some areas do not fully cover internal theoretical detail or the full range of historiographical disagreement.

3. Overview of the Model

The five-stage model describes the process of creation in the following five stages.

Stage	Definition
Field	An undifferentiated state. It refers to an initial condition in which neither direction nor structure has yet been determined.
Wave	A stage of exploration in which multiple directions diverge and differences are amplified.
Edge	A stage of tension in which opposing elements coexist and do not immediately converge on either side.
Vortex	A stage in which a new coherence or order emerges out of tension.
Bundle	A stage in which form is fixed and stabilized as a structure that can be reused.

This model is not a one-way line; it includes a circulation in which Bundle becomes the condition for the next Field. The five stages are also not rigidly separated boxes but are better understood as a continuous spectrum. The model includes the hypothesis that the same structure can repeat at different scales, from momentary thought to institutional formation and long-term historical change.

If a sense of incongruity remains at this point, that is itself an important perspective for the present investigation. Readers may continue while holding on to that incongruity.

4. Investigation Results: Overall Picture

List of Evaluated Theories

Theory	Proponent	Structural Correspondence
Improvisation's information-processing cycle (Pressing)	Pressing (1988)	Strong
The dialectical structure of sonata form	Rosen (1988), Hepokoski & Darcy (2006)	Strong
Emotion generation as expectation, delay, and resolution (Meyer)	Meyer (1956)	Strong
Data-driven research on the universality and diversity of song (NHS)	Mehr et al. (2019), Savage et al. (2015)	Strong
ITPRA theory (Huron)	Huron (2006)	Strong
Groove theory (Pressing/Madison/Witek)	Madison (2006), Witek et al. (2014), Pressing (2002), Janata et al. (2012)	Strong

Theory	Proponent	Structural Correspondence
Tonal Pitch Space (Lerdahl)	Lerdahl (2001), Lerdahl & Jackendoff (1983)	Conditional
The staged structure of raga (Indian classical music)	Powers & Widdess (2001), Jairazbhoy (1971), Clayton (2000)	Conditional
Auditory Scene Analysis (Bregman ASA)	Bregman (1990)	Strong
Ecological music perception (Clarke)	Clarke (2005)	Conditional

Of the 10 evaluated cases, 7 showed strong structural correspondence, 3 showed conditional correspondence, and 0 showed no confirmed correspondence. Overall, this domain exhibits correspondence with the five-stage model at a comparatively high density, but the manner of correspondence is not uniform. Some theories offer a thick description of the conditions of Field, while others describe with precision the tension of Edge or the fixation of Bundle.

What matters most is not the high correspondence rate as such, but which theory describes which stage in what vocabulary. Even when dealing with the same phenomenon of "creation," this domain illuminates different phases of the five stages through different layers, including body, institution, time, perception, practice, and symbol.

5. Investigation Results: Examination of Major Theories

5.1 Improvisation's Information-Processing Cycle (Pressing)

"Improvisation's information-processing cycle (Pressing)," proposed by Pressing (1988), describes improvisation as a seven-step cycle of "event generation -> sensory monitoring -> cognitive representation -> evaluation -> next-action planning -> next action -> feedback" (Pressing, 1988). The difference between improvisation and fixed performance is greatest at the stages of representational evaluation and next-action design. Error is not treated as something to be corrected, but as input into the next act of creation.

Pressing's seven-step cycle displays a high degree of structural correspondence to the five stages as an information-processing version of them. "Cognitive representation and evaluation" corresponds directly to Edge, and the cyclical structure includes the return from Bundle to Field. Particularly notable is that the "creative use of error" in improvisation describes within performance itself the same structure as the stance of taking up rejected error as a question. At the intrapersonal level, it connects with Meyer's "expectation -> delay -> resolution."

The description of "error" in improvisation as input into the next act of creation rather than as an object of correction is, beyond its immediate relevance here, one of the clearest examples of the path in which error is taken up and incorporated. At the same time, Pressing's account is computational and information-processing oriented, and therefore operates at a different descriptive level from creation as ontological emergence, which the five-stage model is originally meant to describe. This difference should not be understood as a difference in kind, but as a difference of scale within the same creative process: third-person description versus first-person description. This belongs to the contexts of music psychology and improvisation studies. Pressing (1988) is foundational, and the model was later developed further in work associated with Berline-Pressing. It also connects with Johnson-Laird's (2002) computational study of jazz improvisation.

Figure 3: Musicology — ×5

5.2 The Dialectical Structure of Sonata Form

"The dialectical structure of sonata form," as discussed by Rosen (1988) and Hepokoski & Darcy (2006), treats the transition in the exposition from tonic to dominant as a "large-scale dissonance" and positions it as a structural problem that ought to be resolved in the recapitulation (Rosen, 1988). Sonata form should be understood not as a fixed template but as a dynamic process. Hepokoski & Darcy (2006) describe the sonata as an "action space."

Sonata form builds the generation of difference, the amplification of relations, and their recovery into form itself. Rosen's "structural problem" is structurally isomorphic with the retention of predictive error as a question. The process in which tonal dissonance intentionally generated in the exposition (Wave) is maximized in the development (Edge) and resolved in the recapitulation (Vortex) can be regarded as an implementation within musical form of Field -> Wave -> Edge -> Vortex -> Bundle.

This belongs to the context of Western music theory. Rosen (1988) and Hepokoski & Darcy (2006) are the two major points of reference. Its applicability to non-Western music is limited, but the mechanism of "intentional tension design" itself may also be observable in the unfolding of raga in Indian classical music, such as alap -> jor -> jhala.

5.3 Emotion Generation as Expectation, Delay, and Resolution (Meyer)

"Emotion generation as expectation, delay, and resolution (Meyer)," proposed by Meyer (1956), argues that musical meaning arises through the generation, inhibition, delay, and resolution of expectation (tendency). Musical tendencies are learned within a particular style system (Meyer, 1956). The inhibition and delay of tendencies produce suspense, and alternative continuations come into view (Meyer, 1956).

Meyer's theory of expectation is the central point of connection in musicology for this project, and correspondence with the five stages is established at high intensity across all stages. The process of "expectation -> delay/inhibition -> suspense -> resolution -> integration of belief" is structurally isomorphic with Field -> Wave -> Edge -> Vortex -> Bundle. In particular, the coexistence of multiple alternative continuations is one of the clearest descriptions of Edge as a space of possibilities, and it provides a direct connection to predictive processing and the free energy principle. The connection here is also clear: expectation violation functions as predictive error held in the form of a question. The precision of tendency corresponds to a threshold of precision, and suspense is a time-art version of retaining tension by delaying reflexive resolution and remaining within uncertainty.

Meyer speaks of emotion as a multicomponent process of expectation, uncertainty, and reevaluation, positioning emotion not as a mere bodily reaction but as the result of a cognitive process. In that respect, his work structurally resonates with Barrett's (2017) constructivist theory of emotion. The five-stage model does not explicitly address emotion, but when emotion is described as something constructed, Meyer's expectation-emotion mechanism provides a concrete case. Huron's (2006) ITPRA theory is a computational and neuroscientific development of Meyer's framework, so there is a clear line of development from Meyer to Huron. This belongs to the contexts of music psychology and musical aesthetics. Meyer (1956) is foundational. Although tonal music in the Western tradition is his main object, Meyer himself states that the framework is applicable to style systems in general. Narmour's (1990) Implication-Realization Model, Huron's (2006) ITPRA theory, and Temperley's (2007) statistical music perception are direct successors to Meyer.

5.4 Data-Driven Research on the Universality and Diversity of Song (NHS)

"Data-driven research on the universality and diversity of song (NHS)," by Mehr et al. (2019) and Savage et al. (2015), reports that musical descriptions are present in 309 of 315 societies in eHRAF. The estimated prevalence of music is close to 100% (Mehr et al., 2019, *Science*, 366). Three dimensions, formality, arousal, and religiosity, explain 26.6% of the variation in song behavior (Mehr et al., 2019).

The research process itself structurally corresponds to the five stages: an undifferentiated phenomenon, song behavior around the world, is turned into data, organized into a small number of axes, condensed into predictability, and stabilized as a reusable resource. This is not a mapping of music's internal process, however, but a mapping of the process of knowledge generation at a metalevel. If the five-stage model describes the creation of being, NHS describes the creation of knowledge, showing that the same five-stage structure can appear at the metalevel as well.

This belongs to the contexts of comparative musicology and evolutionary musicology. It is a large-scale data-driven approach grounded in eHRAF (Human Relations Area Files) and the NHS Discography. It can be regarded as a contemporary successor to Lomax's (1968) Cantometrics project.

5.5 ITPRA Theory (Huron)

"ITPRA theory (Huron)," proposed by Huron (2006), argues that five functionally independent response systems operate in temporal sequence with respect to musical expectation: Imagination (advance imaginative rehearsal), Tension (advance bodily preparation), Prediction (reward for predictive accuracy at the moment of outcome), Reaction (automatic response at the moment of outcome), and Appraisal (conscious post hoc evaluation) (Huron, 2006, *Sweet Anticipation*, MIT Press). The prediction response produces positive emotion when the prediction is accurate. This is a "prediction effect" independent of the pleasantness or unpleasantness of the event itself (Huron, 2006).

In this theory, correspondence with the five stages can be observed through a movement that includes the formation of preconditions, the amplification of difference, the organization of relations, the emergence of a new coherence, and stabilization.

This belongs to the contexts of music psychology and cognitive science. Huron (2006) is a direct successor to Meyer (1956), and also connects with Narmour's (1990) Implication-Realization Model and Temperley's (2007) statistical music perception. The connection to predictive processing is even more direct here than in Meyer, making it a strong candidate for a central cross-domain point of contact.

5.6 Groove Theory (Pressing/Madison/Witek)

"Groove theory (Pressing/Madison/Witek)," as developed by Madison (2006), Witek et al. (2014), Pressing (2002), and Janata et al. (2012), treats groove as the experiential quality of "wanting to move with the music," accompanied by the listener's sensorimotor coupling (Janata et al., 2012, *Journal of Experimental Psychology: General*, 141(1)). There is an inverted-U relationship in which moderate syncopation, that is, intermediate rhythmic complexity, maximizes groove ratings. If the music is too simple or too complex, groove weakens (Witek et al., 2014, *PLoS ONE*, 9(4)).

The generation of groove shows the following structural correspondence to the five stages: 1. **Field (metrical grid)**: equally spaced pulses function as a "sea in which everything is dissolved together." If nothing deviates, groove does not arise. 2. **Wave (syncopation)**: deviations from the grid emerge as sway and opposition. Microtiming, playing ahead of or behind the

beat, generates difference. 3. **Edge (zone of optimal complexity)**: the peak of the inverted U is the boundary between predictability and uncertainty. This is a bodily version of what happens at a boundary. 4. **Vortex (sensorimotor entrainment)**: the body couples with the music and rises as a coherent whole in a self-organizing process. 5. **Bundle (collective synchronization)**: groove endures and becomes stabilized as collective bodily synchrony. The connection is also clear here: deviation from the grid is the difference between prediction and actual sound. The finding that groove is maximized when this difference is "just right," that is, at moderate syncopation, is a bodily version of a threshold of precision.

Groove theory includes the same author, Pressing, as the improvisation model discussed above, but the focus is different. The improvisation model concerns the performer's cognitive cycle, whereas groove theory concerns bodily entrainment between listener and performer. They are different facets of the same phenomenon, musical creation and reception. Its distinctive contribution is that groove is a **bodily** experience. Whereas Meyer and Huron describe the listener's cognition and emotion, groove theory describes response at the level of "the body moves." The fact that Vortex takes concrete form here as body-music coupling offers a perspective not present in the other musicology entries. This belongs to the contexts of music psychology, rhythm research, and ethnomusicology. It originated in the study of rhythmic structures in African diasporic music, including funk, jazz, and Afrobeat, and was later quantified through experimental-psychological methods. Pressing's (2002) "Black Atlantic Rhythm" provides the theoretical basis, while Witek et al. (2014) provide experimental verification. Relative to the improvisation model, Pressing (1988) concerns the cognitive cycle of improvisation, while Pressing (2002) concerns the rhythmic structure of groove. The author is the same, but the object and focus differ, so the two are complementary rather than redundant.

5.7 Tonal Pitch Space (Lerdahl)

"Tonal Pitch Space (Lerdahl)," as developed by Lerdahl (2001) and Lerdahl & Jackendoff (1983), is a model that quantifies the "distance" among pitches, chords, and keys within a multidimensional space. It has a four-layer structure of basic pitch space -> chordal distance -> key distance -> regional distance (Lerdahl, 2001, *Tonal Pitch Space*, Oxford UP). Melodic and harmonic tension can be calculated as distance within the space, providing a mathematical description of musical motion.

In this theory, correspondence with the five stages can be observed through a movement that includes the formation of preconditions, the amplification of difference, the organization of relations, the emergence of a new coherence, and stabilization.

TPS's distinctive contribution is the quantification of tension. Whereas Meyer and Huron describe the qualitative process of expectation and emotion, TPS makes tension calculable as a numerical value. This quantification is a contribution as an analytic tool, however, and it operates at a different descriptive level from the ontogenetic process described by the five-stage model. This belongs to the contexts of music theory and computational musicology. Lerdahl was a central figure in music cognition research at MIT. TPS develops Lerdahl & Jackendoff's (1983) GTTM and also influenced later work such as Rohrmeier's (2011) Temporal Hierarchical Analysis.

5.8 The Staged Structure of Raga (Indian Classical Music)

"The staged structure of raga (Indian classical music)," as discussed by Powers & Widdess (2001), Jairazbhoy (1971), and Clayton (2000), notes that raga performance in Hindustani music has a recognized staged structure: alap (unmetered, unaccompanied exploration of the mode) -> jor (introduction of pulse, gradual acceleration) -> jhala (rapid rhythmic section) -> gat/bandish (fixed composition with tabla accompaniment) (Powers & Widdess, 2001, *New Grove*; Jairazbhoy, 1971). Alap is a free exploration that "drifts through the modal space of the raga" and lacks

metrical structure. The performer gradually discloses the characteristics of the mode, including vadi, samvadi, and arohana/avarohana (Jairazbhoy, 1971).

The staged structure of raga performance shows the following structural correspondence to the five stages: 1. **Field (alap)**: drifting through unmetered modal space is the purest musical realization of a state like a sea in which everything is dissolved together. Western music has few equivalent Field-like sections; the closest is the introduction, though it often remains metrical. 2. **Wave (jor)**: the introduction of pulse corresponds to differences arising within Field and standing forth as oscillation. It is a transition from the unstructured to the structured. 3. **Edge (transitional zone)**: the accelerating zone from jor to jhala partly corresponds to what happens at a boundary, though it may be closer to an amplification of Wave than to Edge proper. 4. **Vortex (gat/bandish)**: the emergence of the fixed composition and the dialogue with the tabla correspond to rising as a coherent whole. 5. **Bundle (completion of performance)**: return to the grammatical rules and tradition of the raga. The greatest contribution here is that the transition from alap through jor to jhala is a continuous process of moving from unstructured to structured, which is structurally similar to the temporal unfolding from Field through Wave and Edge to Vortex.

This belongs to the context of Indian musicology, specifically Hindustani music theory. Jairazbhoy (1971) provides a foundational systematic description of raga, while Clayton (2000) offers an analysis focused on temporal structure. Powers & Widdess (2001) are the standard reference. For comparison with Western music theory, works such as Viswanathan & Allen (2004) *Music in South India* are also relevant.

5.9 Auditory Scene Analysis (Bregman ASA)

"Auditory Scene Analysis (Bregman ASA)," proposed by Bregman (1990), argues that the auditory system decomposes the complex acoustic waves reaching the ear into "auditory streams" corresponding to separate sound sources. This process is called Auditory Scene Analysis (Bregman, 1990, *Auditory Scene Analysis*, MIT Press). Two kinds of process are involved: primitive processes, which are preattentive, automatic, and governed by Gestalt principles, and schema-based processes, which are top-down and grounded in learning (Bregman, 1990).

The perceptual organization process in ASA shows the following structural correspondence to the five stages: 1. **Field (undifferentiated mixture of sound waves)**: the total acoustic signal reaching the ear is itself a sea in which everything is dissolved together. As a continuum of air vibration, it is undifferentiated. 2. **Wave (primitive grouping)**: initial differentiation based on frequency proximity and temporal continuity corresponds to differences arising within Field and standing forth as oscillation. 3. **Edge (competition among groupings)**: the ambiguous auditory stimuli explicitly discussed by Bregman, in which the same signal is perceived bistably as one stream or two, are a perceptual version of what happens at a boundary. The site where multiple grouping interpretations compete is Edge. 4. **Vortex (stream segregation)**: the process by which auditory streams arise as perceptual entities is structurally isomorphic with individuation in the sense of Simondon. 5. **Bundle (perceptual stability)**: the parsed auditory scene stabilizes as a settled perception. The connection here is also clear in an implicit form: prediction in ASA operates prior to conscious awareness. The auditory system carries expectations of how things ought to sound, and the mismatch with actual input drives the revision of grouping. This is a preconscious version of the difference between prediction and reality.

This belongs to the contexts of auditory psychology and perceptual psychology. Bregman (1990) is foundational. Later studies include Winkler et al. (2009), which attempts an integration with predictive processing, and Darwin (2008), which develops computational modeling. Applications to music perception include McAdams & Bregman (1979). Clarke's (2005) ecological music

perception critically inherits ASA while emphasizing the environmental meaning of perception. Where ASA asks how perception is organized, Clarke asks why it is organized in that way.

5.10 Ecological Music Perception (Clarke)

"Ecological music perception (Clarke)," proposed by Clarke (2005), applies Gibson's ecological psychology to music perception. Musical meaning is directly perceived without inference (Clarke, 2005, *Ways of Listening*, Oxford UP). Listeners perceive affordances, that is, possibilities for action, in music. They directly take up what music offers: movement, emotion, and the direction of attention.

Clarke's ecological music perception and the five stages describe the following shared structure in different vocabularies: 1. **organism-environment coupling**: Clarke's theory places the unit of perception in the organism-environment coupled system. This structurally resonates with the fact that the five stages begin from Field, a state like a sea in which everything is dissolved together. Both begin from a condition in which subject and object are not yet sharply differentiated. 2. **detection of invariants**: the process of detecting invariants within environmental structure corresponds to the transition from Field to Wave, the generation of difference. However, Clarke's invariants are understood as already present in the environment, whereas the five-stage model speaks of differences as arising, so the ontological premises differ. 3. **direct perception**: the direct perception of affordances is a non-inferential version of Vortex, the emergence of a coherent whole. The claim that meaning appears without inference resonates with the claim that Vortex is not the result of computational inference but an emergence of being.

Clarke's greatest distinctiveness lies in his **critique of information-processing approaches**. Bregman ASA, Pressing, and Huron all stand within information-processing frameworks. Clarke criticizes these approaches for reducing perception to internal processing and instead describes perception as an ecological relation between organism and environment. This belongs to the application of ecological psychology, derived from Gibson (1979), to music perception. Clarke (2005) is the representative work, and later developments include DenPys (2012) on the ecological dynamics of music performance. It should be read in contrast with Bregman ASA.

6. Cross-Domain Patterns

First, as shown by Improvisation's information-processing cycle (Pressing), the dialectical structure of sonata form, and emotion generation as expectation, delay, and resolution (Meyer), this domain describes difference and conflict not merely as obstacles but as conditions for the emergence of a new order. This means that, within the five-stage model, the transition from Edge to Vortex is especially richly observable in this domain.

Second, what data-driven research on the universality and diversity of song (NHS), ITPRA theory (Huron), and groove theory (Pressing/Madison/Witek) share is that generation is not a one-off event but includes a circulation through the formation of preconditions, the amplification of tension, local ordering, stabilization, and subsequent reconfiguration. This confirms that the five-stage model must be read not as a static table of classification but as a temporal movement.

On the other hand, Tonal Pitch Space (Lerdahl), the staged structure of raga (Indian classical music), and ecological music perception (Clarke) also show that correspondence with the five-stage model does not arise automatically. Similarity in the number of stages or in vocabulary is insufficient; it is necessary to examine in each case which concept bears the turning point and at what level the structure emerges.

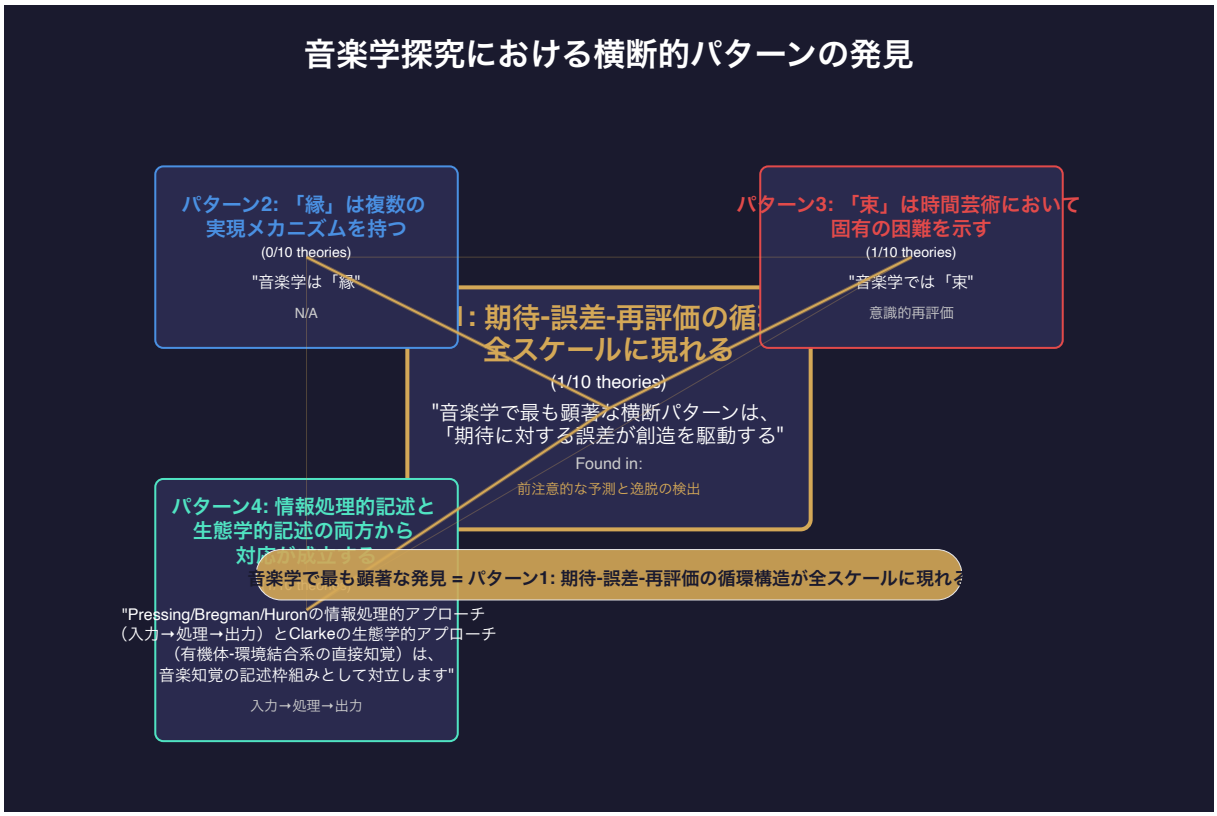

Figure 4: Musicology —

7. Unresolved Questions

1. To what extent can Tonal Pitch Space (Lerdahl), the staged structure of raga (Indian classical music), and ecological music perception (Clarke) be read as stage-based descriptions within the five-stage model?

2. How far can the structures confirmed in this domain be extended from the explanation of individual cases into a more general generative model? The conditions under which comparability can be maintained without losing the specificity of the objects still require further scrutiny.

3. Should the five stages be read as a linear progression, or as a circulation that includes multiple temporal layers and repetition? In particular, theories differ in the degree to which they make explicit the return from Bundle to the next Field.

4. What becomes visible, and what is lost, when the vocabulary proper to this domain is translated into the general vocabulary of the five stages? The boundary between translatability and untranslatability is itself an important issue for future comparative inquiry.

8. Conclusion

This re-examination of the domain confirmed strong structural correspondence in 7 of the 10 theories and conditional correspondence in 3. In particular, Improvisation's information-processing cycle (Pressing), the dialectical structure of sonata form, and emotion generation as expectation, delay, and resolution (Meyer) provide especially thick descriptions either from Field to Edge or from Edge to Vortex, and they therefore count as central evidence within this domain.

At the same time, the findings of this domain do not support the claim that the same five stages

simply appear in identical form in every theory. The stages that appear most strongly differ from one theory to another, and some cases require reservations in stage mapping. What matters, therefore, is not to generalize directly from a high correspondence rate, but to continue comparing which phase is described in which vocabulary.

In that sense, this domain not only supports the validity of the five-stage model in one direction but also imposes conditions on how the model should be read. The very fact that the domain contains both theories with thick correspondence and theories that require reservation constitutes part of its value as a comparative investigation.